

Blockchain-Assisted Secure Academic Learning Record Management

Namrata Mariam Chacko, Anvesha Singh, Vidya Kamath

Manipal Institute of Technology

Manipal Academy of Higher Education, Manipal, Karnataka, India

namrata.c@manipal.edu, anvesha.mitmpl2024@learner.manipal.edu, kamath.vidya@manipal.edu

Abstract—Academic Learning Record Management (ALRM) systems remain largely centralized, creating risks related to credential fraud, and inefficient verification processes. This paper proposes a Blockchain-Assisted Self-Sovereign Identity (Ba-SSI) enabled ALRM framework that transforms the traditional Academic Bank of Credits (ABC) into a role-governed, cryptographically verifiable academic identity infrastructure. The system employs commitment-based credential anchoring on an Ethereum Virtual Machine (EVM)-compatible blockchain, combined with off-chain artifact storage and smart contract-enforced lifecycle control for issuance, attestation, revocation, and verification.

An EVM smart contract prototype was implemented and deployed in a permissioned environment (Kaleido), incorporating issuer accreditation, authority-based attestation, and status-based revocation mechanisms. Performance evaluation was conducted under varying concurrency levels (1–20) with transaction loads up to 100 operations. Experimental results demonstrate stable scalability, with throughput increasing from 0.20 TPS at single concurrency to 1.51 TPS at 20 concurrent users. Average transaction latency decreased proportionally with concurrency, while gas consumption remained predictable and bounded per credential anchoring operation.

Index Terms—Higher Education, Blockchain, Self-Sovereign Identity, Academic Records, Learning Record Management, Academic Bank of Credits, Certificate Verification, Decentralization

I. INTRODUCTION

Certificates obtained upon completing an educational course are crucial for pursuing higher education or for applying for a job. However, numerous events shed light on how fraudsters have forged certificates [1]. Multiple cases have been reported pan India, in one case, the accused revealed that they had sold at least 50,000 fake degrees and mark sheets of various universities and school boards [2], [3]. Currently, the authentication is based on holographic stickers, which can easily be replicated. Employers also have a daunting task to verify the certificates they receive. Companies need to invest significant time and money in manually verifying applicants' certificates. While existing e-certificates provide a minimal degree of security, if the signing central authority is compromised, the entire certificate data will be available to hackers [4], [5]. Besides fake certificates, there are also universities that are not registered with UGC [6], [7].

According to the article published by Dr. N Prabhudeva on Aug 17, 2019. [8], there has been a surge in diploma mills and fake certificates, which some individuals use to secure jobs worldwide. The current certification issuing system in India is

via hard copy. With the advent of online courses, students and employers find it difficult to identify genuine universities or certificates. Blockchain is a disruptive technology that enables the immutable recording of data on a distributed ledger. The blockchain technology can help to maintain the authenticity, integrity and security of certificates. It can provide a decentralized framework to share the certificate with potential employers which are easily verifiable. When looking for jobs abroad, graduates need to get their certificates attested from the Ministry of Human resource department, which can take up to months to be processed, Blockchain technology can speed up the process to a few days. A change to an Academic Learning Record Management (ALRM) system that securely documents academic credentials (AC) and achievements (LRM) is necessary. There should be a new computerized approach that standardizes and streamlines ALRM processes across the globe. Research seeks improved security, integrity, and authenticity by describing how Blockchain can efficiently prevent certificate forgery and help achieve a single digital platform [4], [9], [10].

A pragmatic research paradigm will allow for both quantitative and qualitative proofs to examine if the introduction of Blockchain technology will be a feasible solution to guarantee verifiable, genuine diplomas. As it can provide an ontological stance of the socio economic impact along with an experimental approach to prove empirically the design of a secure certificate generation system. The research will be able to help in constructing and analyzing the role of Blockchain in shaping the future of the certification system.

A. Contributions

This article proposes a blockchain-based framework for the verification of educational certificates. Along with a case study on the impact of a blockchain-based certification initiative. The major contributions are :

- Implementation of the certificate storage and verification system.
- Empirical analysis of the system, based on cost and network latency.

B. The Technological Aspects of Blockchain

Blockchain refers to a distributed ledger system that records transactions securely and immutably.

- **Structure:** Each transaction is recorded in a block linked to the previous block by including the previous block's hash. This ensures tamper-resistance.
- **Elements:** A block typically includes a block header, Merkle tree root (for integrity), transaction data, and metadata.
- **Nodes:** Full nodes store a complete copy of the ledger and validate all transactions, while light nodes verify details via full nodes.
- **Smart Contracts:** Event-condition-action (ECA) programs that automatically execute agreements between parties without requiring mutual trust.

II. BLOCKCHAIN-BASED ALRM APPLICATIONS: COMPARATIVE ANALYSIS

Blockchain has been explored across multiple dimensions of Academic Learning Record Management (ALRM), including credential issuance, revocation, credit transfer, verification, and lifelong learning record management. Table I consolidates representative systems and highlights architectural choices and limitations.

Across these systems, recurring structural limitations emerge: (i) over-reliance on public Proof-of-Work networks leading to high cost and low throughput, (ii) centralized revocation endpoints or certificate authority dependencies, (iii) inefficient on-chain storage of full artifacts rather than cryptographic commitments, and (iv) limited integration of decentralized identity and governance models. These gaps motivate a role-governed, commitment-based anchoring architecture with distributed off-chain storage and formal performance evaluation, as proposed in this work.

III. PROPOSED SECURE ACADEMIC LEARNING RECORD MANAGEMENT

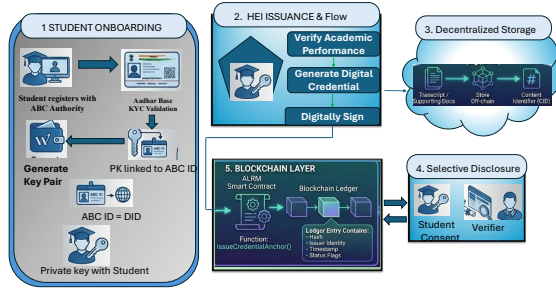


Fig. 1: System Architecture for the Proposed Learning Management System

The proposed framework integrates Blockchain-Assisted Self-Sovereign Identity (Ba-SSI) into the Academic Learning Record Management (ALRM) model, transforming the traditional Academic Bank of Credits (ABC) into a decentralized, cryptographically verifiable ecosystem (Fig. 1). The system operates as a consortium network comprising Higher Education Institutions (HEIs), the ABC Authority/Ministry

of Education (MoE), learners, and external verifiers, governed through role-based smart contract logic and auditable blockchain transactions.

The lifecycle begins with learner onboarding. After KYC validation through the ABC Authority, the student generates a cryptographic key pair within an SSI wallet. The public key is linked to the ABC ID, effectively converting it into a decentralized identifier (DID), while the private key remains solely under learner control, establishing the identity layer of the system.

Upon course completion, the HEI verifies academic performance and issues a digitally signed credential containing course metadata and credits. The full transcript is stored off-chain (e.g., IPFS), generating a content identifier (CID). A cryptographic commitment (hash of the credential) is then anchored on-chain via the `issueCredentialAnchor()` function, storing only minimal metadata (hash, issuer, timestamp, status flags) to ensure privacy and integrity.

Governance is enforced through an attestation phase, where the ABC Authority validates institutional accreditation and records endorsement using `attestCredential()`. This creates immutable proof of both issuance and regulatory approval. Credits accumulate on-chain through DID-linked anchors, enabling portability across institutions without centralized databases.

For disclosure, the learner generates a Verifiable Presentation (VP) via the SSI wallet, revealing only required attributes. The verifier performs multi-layer validation: (i) signature verification, (ii) on-chain anchor confirmation, (iii) revocation status check, and (iv) optional IPFS hash validation.

If required, authorized entities invoke `revokeCredential()` to update credential status without altering historical records, ensuring auditability and integrity.

Overall, the Ba-SSI-enabled ALRM framework establishes a decentralized academic identity infrastructure combining on-chain commitment anchoring, distributed storage, selective disclosure, and role-governed lifecycle management to achieve immutability, portability, and regulatory compliance.

IV. PERFORMANCE ANALYSIS

Table II presents a comparative analysis of the proposed Academic Learning Record Management (ALRM) framework against existing blockchain-based academic record systems across key security, privacy, and performance dimensions. Unlike prior works, the proposed ALRM integrates distributed off-chain storage, formal cost and execution-time evaluation, and scalability analysis while maintaining confidentiality, authenticity, integrity, and privacy through a role-governed blockchain architecture.

A. Experimental Setup

To validate the feasibility and scalability of the proposed Academic Learning Record Management (ALRM) smart contract, we conducted benchmarking on an Ethereum Virtual Machine (EVM)-compatible permissioned network provisioned

TABLE I: Comparative Analysis of Blockchain-Based Academic Record Systems

Category	Representative Systems	Core Features	Key Limitations
Credential Issuance	Accredible [10], Blockcerts [11], CVSS [12], Cerberus [13], NFT-based models [14]	Hash anchoring on Bitcoin/Ethereum; smart contracts; Merkle batching; QR/NFT credentials	High PoW costs; centralized revocation endpoints; one-contract-per-student inefficiency; weak identity binding
Revocation	BTCert [15], BlockCerts [16], IU-TransCert [17], Cerberus [13]	Multi-signature revocation; Merkle audit trails; on-chain policy enforcement	Separate on-chain transactions per revocation; scalability constraints; CA dependence; address management overhead
Credit Transfer	EduCTX [18], Srivastava et al. [19], Manoj et al. [20], ABC models [21]	Consortium blockchains; DPoS; multi-signature transfers; Ethereum smart contracts	Limited accreditation integration; centralization risk; costly on-chain storage; conceptual-stage implementations
Verification Platforms	CHESICC, Gradbase, Docschain, Delgado et al.	QR-based verification; consortium sidechains; hybrid storage models	Centralization risks; expensive on-chain storage; OCR overhead; interoperability challenges
Learning Record Management	Blockchain+Kudos [22], BOLL [23], UniChain [24], EduChain [25]	Student-centric EARs; hybrid blockchain models; PoA/PoW architectures	Plaintext storage; governance gaps; GDPR ambiguity; scalability constraints

TABLE II: Comparison of the Proposed Academic Learning Record Management (ALRM) Framework with Existing Blockchain-Based Academic Record Systems

Reference	Blockchain Type	Confidentiality	Authenticity	Integrity	Privacy	Off-chain Storage	Cost Analysis	Execution Time Evaluation	Scalability Analysis
[26]	Permissioned	✗	✓	✓	✓	✗	✗	✗	✗
[27]	Permissionless	✗	✓	✓	✗	✗	✗	✗	✗
[28]	Permissioned	✓	✓	✓	✗	✗	✗	✗	✗
[29]	Permissioned	✓	✓	✓	✓	✓	✓	✗	✓
Proposed ALRM	Permissioned	✓	✓	✓	✓	✓ (IPFS)	✓	✓	✓

via Kaleido. The deployed `ALRMRegistry` contract implements (i) role-based governance (Owner, Issuer, Attester), (ii) credential anchoring via cryptographic commitments, (iii) authority-based attestation, (iv) controlled revocation, and (v) lightweight verification primitives. All experiments were executed using Hardhat-based scripts that submit transactions through authenticated JSON-RPC endpoints and measure on-chain resource usage and end-to-end confirmation latency.

We evaluate three classes of metrics: (1) *gas consumption* per operation, (2) *transaction latency* from submission to confirmation, and (3) *throughput* measured as transactions per second (TPS) under varying concurrency levels. For scalability, we executed batches with concurrency levels $\{1, 5, 10, 20\}$ and total transaction counts of 50 and 100 to observe saturation behavior under increasing load.

a) Statistical Reporting.: For each metric, we report the sample mean and sample variance. Given observations $\{x_i\}_{i=1}^n$, the mean μ and variance σ^2 are computed as:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i, \quad \sigma^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2. \quad (1)$$

Gas measurements are collected from transaction receipts (`gasUsed`), while latency is measured in milliseconds as wall-clock elapsed time between submission and confirmation.

Throughput is computed as:

$$\text{TPS} = \frac{N}{T}, \quad (2)$$

where N is the number of successful transactions and T is the total elapsed execution time in seconds for the batch.

B. Gas Consumption Analysis

Figure 2a summarizes the gas cost profile across the core credential lifecycle operations. Credential anchoring incurs the highest gas usage because it stores the commitment and metadata and emits an anchoring event, while attestation and revocation primarily update state flags and emit corresponding events. Importantly, gas usage remains stable across repeated executions (low empirical variance), indicating that the contract's computational path is largely deterministic and insensitive to input size due to the design choice of storing only fixed-length hashes and minimal pointers on-chain. This supports the design goal of minimizing on-chain footprint by externalizing transcript artifacts to off-chain storage while anchoring integrity commitments on-chain.

a) Interpretation with Statistics.: Let $g_{op,i}$ denote the gas used by operation op on trial i . The sample mean $\mu_{g,op}$ and variance $\sigma_{g,op}^2$ (Eq. 1) quantify computational stability. In our experiments, the observed low dispersion across trials indicates that the contract's execution cost is dominated by fixed

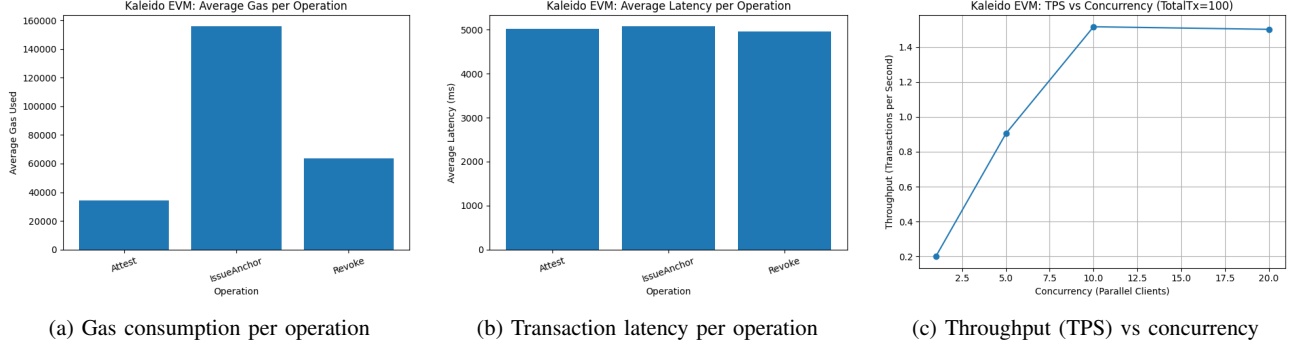


Fig. 2: Performance evaluation of the proposed ALRM smart contract on an EVM-compatible permissioned network. (a) Gas consumption highlights on-chain cost per credential lifecycle operation. (b) Confirmation latency reflects network processing and block finality. (c) TPS increases with concurrency until network saturation.

Algorithm 1 ALRMRegistry

Input: credId, holderIdHash, commitment, expectedCommitment

Output: (match, attested, revoked, issuer)

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1: Initialisation: Maintain issuer registry and anchor mapping.
2: IssueAnchor
3: if Caller has ISSUER_ROLE and credId is new then
4:   Store (issuer, holderIdHash, commitment, issuedAt)
5: end if
6: Attest
7: if Caller has ATTESTER_ROLE and anchor exists and not revoked then
8:   Set attested  $\leftarrow$  true
9: end if
10: Revoke
11: if Caller is issuer or Caller has ATTESTER_ROLE then
12:   Set revoked  $\leftarrow$  true
13: end if
14: Verify
15: if anchor exists then
16:   match  $\leftarrow$  (storedCommitment = expectedCommitment)

17: return (match, attested, revoked, issuer)
18: else
19:   return (false, false, false, null)
20: end if

```

storage and event emission costs rather than data-dependent computation.

C. Latency Analysis

Transaction latency captures the user-observed delay between submitting an operation (e.g., anchoring or revocation) and receiving confirmation. Figure 2b reports average confirmation time per operation. The observed latency is relatively uniform across operations, suggesting that network-level scheduling and block finality dominate end-to-end delay rather

than smart-contract instruction complexity. This is consistent with EVM execution where per-transaction compute is small compared to block production and inclusion time.

a) Interpretation with Statistics.: Let $\ell_{op,i}$ denote the confirmation latency for operation op on trial i . The mean $\mu_{\ell,op}$ and variance $\sigma_{\ell,op}^2$ (Eq. 1) characterize both typical delay and jitter. Across our runs, latency variance is primarily attributable to transient network conditions and block inclusion timing; the contract itself contributes minimal variability because it performs bounded, fixed-size state updates.

D. Throughput and Scalability

Throughput is evaluated by increasing the number of concurrent transaction submissions and measuring aggregate completion rate. Figure 2c shows that throughput increases from ≈ 0.20 TPS under sequential execution (concurrency = 1) to a peak of ≈ 1.52 TPS around concurrency = 10 for the 100-transaction workload, after which throughput saturates (concurrency = 20 yields ≈ 1.50 TPS). This plateau indicates the onset of network saturation, where the limiting factor is block production/consensus throughput rather than contract compute. In other words, beyond a certain concurrency, additional parallel submission increases contention for block space and does not translate into proportional throughput gains.

a) Stability Across Workloads.: The 100-transaction workload exhibits more stable throughput estimates than smaller workloads due to reduced sensitivity to startup overheads and transient confirmation jitter.

b) Optional Variance Reporting for TPS: If multiple runs are performed per concurrency level, TPS variance can be reported as $\sigma_{TPS,c}^2$ computed over repeated trials (Eq. 1). This provides confidence intervals and improves reproducibility for journal submission.

E. Discussion and Implications

Overall, the results validate that the proposed ALRM contract achieves predictable on-chain cost and stable performance under load. The gas profile confirms that the anchoring-based design remains efficient because it persists only fixed-size

commitments and minimal metadata rather than full academic documents. Latency measurements indicate that confirmation time is governed mainly by network-level inclusion and finality, making the contract suitable for consortium settings where block parameters can be tuned. The throughput experiments demonstrate typical EVM scaling behavior: increasing parallelism improves utilization up to a saturation point, beyond which the consensus and block processing pipeline become the dominant bottleneck.

a) Limitations and Future Evaluation.: This evaluation focuses on the on-chain anchoring and governance layer. End-to-end Ba-SSI features (e.g., selective disclosure with zero-knowledge proofs and wallet-integrated verifiable presentations) are executed off-chain and are not reflected in the on-chain gas/latency figures. Future work will (i) integrate a full verifiable credential pipeline and measure end-to-end verification latency, (ii) evaluate distributed storage availability and retrieval performance (e.g., IPFS pinning strategies), and (iii) extend benchmarking across Hyperledger Fabric and Corda to quantify consensus-driven trade-offs in throughput, latency, and operational cost. Although integrating blockchain into ALRM systems offers many advantages, several technological, organizational, and regulatory obstacles remain. This section discusses benefits, problems, and mitigation strategies.

V. DEPLOYMENT IMPLICATIONS AND SYSTEM CONSIDERATIONS

The proposed ALRM framework leverages blockchain primarily as a trust anchor rather than a data warehouse. By anchoring cryptographic commitments of credentials on-chain and storing artifacts off-chain (e.g., IPFS), the system preserves integrity and auditability while minimizing storage overhead and privacy exposure.

Role-based governance enforced through smart contracts enables deterministic lifecycle control over credential issuance, attestation, and revocation. Unlike centralized academic repositories, verification is performed independently through signature validation and on-chain commitment matching, eliminating reliance on intermediary databases. The integration of SSI principles further shifts credential control to the learner while maintaining institutional and regulatory oversight.

A. Technical Constraints and Mitigation Strategies

Despite demonstrated stability in gas consumption and throughput under a permissioned EVM environment, broader deployment introduces technical considerations summarized in Table III. These constraints relate to scalability, identity infrastructure maturity, privacy guarantees, and interoperability across heterogeneous blockchain platforms.

Overall, the current implementation validates the feasibility of a role-governed anchor registry for academic credential management, while highlighting scalability optimization, full SSI integration, and cross-platform benchmarking as primary directions for system evolution.

TABLE III: Technical Constraints and Mitigation Strategies

Constraint	Impact on ALRM	Mitigation Approach
Scalability and Cost	Public EVM networks exhibit gas volatility and limited TPS under high concurrency.	Permissioned deployment (PoA), batching, Layer-2 rollups, and selective on-chain anchoring.
Identity Infrastructure	Full W3C DID/VC stack not yet integrated in prototype.	Future SSI wallet integration and verifiable presentation support.
Immutability vs. Revocation	Permanent ledger entries conflict with dynamic credential invalidation.	Status-based revocation flags without altering historical anchors.
Data Privacy	On-chain data visibility may expose sensitive metadata.	Store only hashes on-chain; encrypt off-chain artifacts; apply selective disclosure mechanisms.
Interoperability	Cross-platform academic portability requires multi-chain compatibility.	Standard adoption (W3C DID/VC) and benchmarking across Fabric and Corda.

VI. IMPLEMENTATION SCOPE AND FUTURE WORK

The ALRM framework comprises governance, identity management, credential lifecycle control, distributed storage, and performance evaluation layers. The present deployment implements the on-chain governance and credential lifecycle modules through an EVM-compatible smart contract. Specifically, the prototype includes: (i) role-based access control (Owner, Issuer, Attester), (ii) issuer accreditation registry, (iii) credential anchoring via cryptographic commitments, (iv) authority-based attestation, (v) controlled revocation, and (vi) lightweight on-chain verification of credential status.

A comprehensive performance evaluation was conducted, covering gas consumption, transaction latency, and throughput (TPS) scalability under varying concurrency levels. These components establish the core trust, integrity, and governance mechanisms of the ALRM ecosystem.

However, several architectural layers remain beyond the current prototype scope. A fully W3C compliant DID registry and Verifiable Credential (VC) issuance stack are not yet implemented; learner identity is presently represented by hashed identifiers, with selective disclosure assumed off-chain. Future work will integrate SSI wallets, verifiable presentations, and zero-knowledge proof-based disclosure mechanisms.

Similarly, while the contract stores content-addressable pointers (e.g., IPFS CIDs), automated storage orchestration, artifact encryption, and availability guarantees are not yet realized. Planned extensions include secure IPFS deployment, distributed pinning, and encrypted artifact management.

Finally, advanced governance and interoperability features—such as consortium voting, automated cross-institution credit reconciliation, multi-entry/multi-exit accumulation logic, and benchmarking across alternative platforms (e.g., Hyperledger Fabric and Corda)—remain open research directions.

VII. CONCLUSION

This work presented a Blockchain-Assisted Self-Sovereign Identity (Ba-SSI) enabled Academic Learning Record Man-

agement (ALRM) framework that transforms the traditional Academic Bank of Credits (ABC) model into a role-governed, cryptographically verifiable academic credential infrastructure. By anchoring credential commitments on-chain while storing artifacts off-chain, the proposed architecture achieves integrity, auditability, and portability without exposing sensitive academic data. Experimental evaluation across varying concurrency levels confirms predictable gas consumption and stable throughput in a permissioned deployment environment.

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